

Steam Turbine Test Rig — 3D Digital Twin Model

Project Context: Educational digital-twin visualization platform for engineering research and turbomachinery testing.

1. 3D Model Geometry

Property	Value
Vertices	1,698
Faces	1,596
File Format	glTF 2.0 Binary (.glb)
Total File Size	~50 KB
Model Status	Prototype — basic geometric representation

2. Component Assembly

The 3D model comprises 15 mesh objects representing the major turbine system components:

Component	Meshes	Description
Steam Admission	4	Triple-valve inlet arrangement (ESV, TV1, TV2, pilot lines)
Turbine Casing	3	
Gearbox	1	Integral speed-reduction gearbox
Dynamometer	1	Load absorption/power measurement unit
Exhaust	1	Low-pressure exit manifold
Leakage Lines	4	Wheel case and bearing drain/leakage collection points

3. Material Properties (PBR)

All components use physically-based rendering (PBR) with metallic + roughness workflow:

Component	Base Color (sRGB)	Metallic	Roughness
Steam Admission Valves	(0.45–0.70, 0.45–0.55, 0.40–0.60)	0.7	0.35
Turbine Casing	(0.35–0.50, 0.45–0.52, 0.52–0.60)	0.7	0.35
Gearbox Housing	(0.50, 0.50, 0.52)	0.7	0.35
Dynamometer	(0.38, 0.42, 0.46)	0.7	0.35
Exhaust Manifold	(0.32, 0.35, 0.38)	0.7	0.35
Leakage/Drain Lines	(0.60, 0.60, 0.65)	0.7	0.35

4. Instrumentation Architecture

Overall Performance Channels (Page 1):

Subsystem	Parameter	Sensor ID	Unit
Inlet Steam	Pressure	PT109	kg/cm²
	Temperature	TT109	°C
	Mass Flow	FT110	TPH
ESV Actuator	Pressure Command	PT108	kg/cm²
TV1	Pressure	PT111	kg/cm²
	Temperature	TT111	°C
TV2	Pressure	PT112	kg/cm²
	Temperature	TT112	°C
Gearbox	Torque Output	GB_TRO	N.m
Turbine	Inlet Temp	PYRO_T	°C
Dynamometer	Power Output	—	kW
	Speed	—	RPM
	Torque	—	N.m
Wheel Case	Pressure	PT120	kg/cm²
	Temperature	TT120	°C
	Flow (Leakage)	FT162	TPH
Inter-GBC	Pressure	PT161/PT163	kg/cm²
	Temperature	TT161/TT163	°C
Exhaust	Pressure	PT150Y	kg/cm²
	Temperature	TT150Y	°C
Leakage Lines	Pressure (L1–L3)	PT162, PT161, PT163	kg/cm²

Turbovisory Vibration Monitoring (Page 2):

Location	Direction	Type	Sensor ID
Top Turbine Casing	Axial	Vibration	ZT600
Bottom Turbine Casing	Axial	Vibration	ZT601
Turbine Shaft (Inlet)	X-Radial	Vibration	XT600
Turbine Shaft (Inlet)	Y-Radial	Vibration	XT601
Turbine Shaft (Outlet)	X-Radial	Vibration	XT602
Turbine Shaft (Outlet)	Y-Radial	Vibration	XT603
Gearbox (Inlet Side)	X-Radial	Vibration	XT604
Gearbox (Inlet Side)	Y-Radial	Vibration	XT605
Gearbox (Outlet Side)	X-Radial	Vibration	XT606
Gearbox (Outlet Side)	Y-Radial	Vibration	XT607

5. Current Model Limitations

- Geometry is highly simplified — basic box/cylinder primitives without realistic detail
- No shaft rotation capability or bearing representations
- Valve assemblies lack mechanical detail and valve trim
- Gearbox lacks individual gear mesh geometry
- No piping, manifold detail, or instrumentation probe representations
- Materials use flat colors; no textures or surface detail
- Model scale and proportions are approximate only
- Casing openings and cutaway views not modeled
- Not suitable for high-zoom or close-up inspection views
- Does not represent industrial appearance or reality

6. Model Accuracy & Authenticity

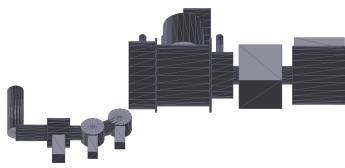
Geometry Accuracy: The current GLB is a placeholder representation. Exact dimensions are unknown; proportions are estimated from the schematic layout. The model does NOT represent a specific Triveni Turbines product.

Turbine Class: Based on the schematic, this appears to be a small single-stage steam turbine with triple-throttle valve admission, likely in the 100–500 kW range. Exact specifications are unavailable in the provided documentation.

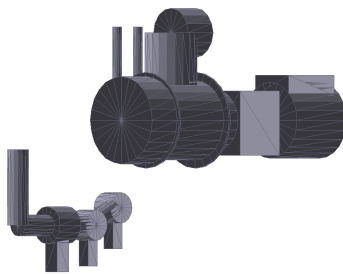
7. Project Context

This is an **educational research visualization project** for a digital-twin dashboard built with ThingsBoard + Babylon.js. The 3D model serves as a visual reference for a real turbine test rig; it is not used for mechanical analysis, design, or commissioning.

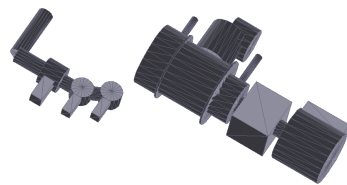
8. 3D Model Renderings



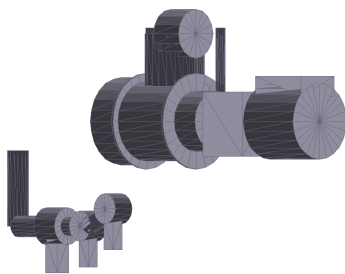
Front View



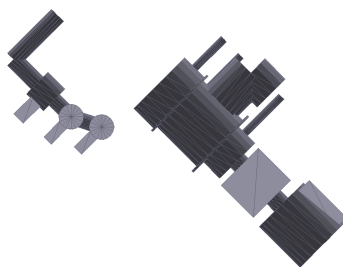
Right View



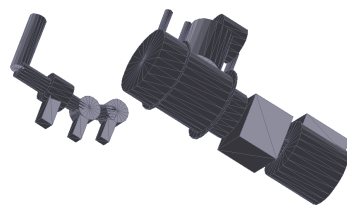
Rear View



Left View



Top View



Isometric View

9. Recommendations for Perplexity Analysis

This specification and the accompanying screenshots are provided for Perplexity to review the current 3D model state and identify improvement opportunities. Key questions for analysis:

- What specific improvements would make this model more visually realistic for a professional digital-twin dashboard?
- Are there publicly available CAD models or references for similar small steam turbines that could inform model refinement?
- What level of geometric detail is necessary for an interactive web-based 3D viewer (Babylon.js)?
- What material/texture improvements would most effectively increase visual fidelity with minimal polygon count increase?
- How should instrumentation probes and sensor positions be represented in the 3D model?

Document generated for educational research purposes. This project is not affiliated with Triveni Turbines or any turbine manufacturer. 3D model is a simplified representation only.